find open source dataset for crime prediction in hull uk. we wnna fit geosmile to it

Worked for 2m 41s

Best primary dataset: Police.uk street-level crime data



The strongest open dataset for fitting **GeoSMILE to crime prediction in Hull** is the official **Police.uk Street-Level Crime and ASB dataset**.

Select:

- **Force:** Humberside Police
- **Period:** July 2023–June 2026
- **Dataset:** Crime data
- **Area filter:** Kingston upon Hull, local-authority code `E06000010`

The downloadable CSV files contain:

- month
- anonymised longitude and latitude
- 2021 LSOA code and name
- crime category
- last recorded outcome
- persistent hashed crime ID
- approximate street/location description

The dataset is updated monthly and published under the **Open Government Licence v3.0**. The current published coverage is July 2023 through June 2026.

Data Poli... +1

Hull currently has **168 LSOAs**, giving approximately:

$$168 \text{ LSOAs} \times 36 \text{ months} = 6,048 \text{ LSOA-month observations}$$

before creating separate crime-category records. That is a reasonable sample for an initial XGBoost–GeoSMILE experiment. Hull Data Ob...

Important spatial limitation

Police.uk coordinates are **not exact offence locations**. Each incident is moved to a nearby anonymisation point, and the published LSOA corresponds to that anonymised point. Police.uk explicitly warns that the coordinates represent approximate locations rather than the place where the crime occurred. Therefore, I would aggregate the incidents to **LSOA-month**, rather than treating them as accurate point-level observations. Data Polic...

Recommended supporting datasets

Dataset	Variables for GeoSMILE	Role
ONS Census 2021	population density, age groups, unemployment, occupation, education, housing tenure, household composition, car availability	Socio-demographic branch
English Indices of Deprivation	income, employment, education, health, housing barriers and living-environment domains	Neighbourhood disadvantage
ONS 2021 LSOA boundaries	polygons, centroids, area, neighbours	Geo branch and mapping
OpenStreetMap	pubs, nightlife, retail, transport stops, parks, schools, roads, intersections and land uses	Built-environment features
DfT road-traffic data	traffic counts and road activity	Ambient population/accessibility
Met Office data	rainfall, temperature and extreme-weather indicators	Monthly temporal covariates

Census 2021 bulk files are available at LSOA level, and the ONS API is unrestricted and does not require an API key. Nomis ... +1

The 2025 deprivation release provides ranks, scores, deciles, underlying indicators and LSOA spatial datasets. However, its overall IMD score incorporates a **crime domain**. We should therefore exclude both the overall

IMD score and the crime domain when predicting crime, because they could introduce target leakage. The remaining domains can be used separately. GOV... +1

OpenStreetMap data are available under the ODbL licence and can provide POI and road-network variables. OpenStree... +1 DfT road-traffic data are openly available through downloads and a REST API, although individual road-link estimates are less robust than regional statistics. Road Traffic ... +1

Best prediction formulation for GeoSMILE

I would **not** begin with event-level crime-type classification. A more defensible GeoSMILE task is:

$$Y_{i,t+1} = \text{number of crimes in LSOA } i \text{ during the following month}$$

Possible targets are:

1. total crime count;
2. violence and sexual-offence count;
3. burglary count;
4. shoplifting count;
5. binary hotspot status, such as whether the LSOA is in the highest 20% next month.

Start with **total monthly crime count**. It produces a continuous or count response that is easier to explain spatially, avoids severe crime-category imbalance, and supports tract-level GeoSMILE narratives.

Suggested row structure

Each row represents one `LSOA × month`:

```
lsoa21cd
month
centroid_x
centroid_y
population_density
young_adult_percentage
unemployment_percentage
social_renting_percentage
single_person_households
income_deprivation_score
employment_deprivation_score
education_deprivation_score
pub_density
retail_density
bus_stop_density
```

```
road_length_density
intersection_density
crime_lag_1
crime_lag_2
crime_lag_3
crime_rolling_mean_3
crime_rolling_mean_6
month_sin
month_cos
target_next_month_crime_count
```

For the target, use either:

```python
objective="count:poisson"
```

in XGBoost, or apply:

```python
np.log1p(crime_count)
```

and fit a standard regression model.

GeoSMILE branch configuration

Geo branch

Use only spatial information:

```
LSOA population-weighted centroid easting
LSOA population-weighted centroid northing
optional neighbouring-LSOA structure
```

British National Grid coordinates would be preferable to raw latitude and longitude because distances are represented in metres.

Feature-A branch

Use all substantive explanatory variables but exclude:

```
coordinates
LSOA code
LSOA name
ward name
geometry
```

This provides the genuinely non-spatial model.

Feature-B branch

Use:

```
all Feature-A variables
+
centroid easting
centroid northing
```

This provides the location-inclusive benchmark and permits comparison with your existing GeoSMILE architecture.

Temporal split

Do not randomly split rows, because adjacent months from the same LSOA would leak highly similar observations across training and testing.

A suitable initial split is:

```
Training:   July 2023 – June 2025
Validation: July 2025 – December 2025
Testing:    January 2026 – June 2026
```

For next-month prediction, the final usable predictor month would be May 2026 because its target is June 2026.

A second evaluation should hold out geographically contiguous groups of LSOAs. That will tell us whether the model genuinely transfers spatially rather than memorising Hull neighbourhoods.

My recommendation

Use this dataset stack:

```
Police.uk crime events
        ↓
aggregate to 168 Hull LSOAs by month
        ↓
join Census 2021 variables
        ↓
join non-crime deprivation domains
        ↓
calculate OSM built-environment indicators
        ↓
create lagged crime and seasonal features
        ↓
train XGBoost count-regression model
        ↓
```

```
apply GeoSMILE
        ↓
aggregate local explanations by LSOA and over time
```

The Hull Data Observatory can be used as an additional validation and profiling source: it provides downloadable data by theme and geography, including LSOA profiles, and its crime series had been updated through March 2026 at the time of checking. Police.uk remains the better primary source because it provides incident-level records and currently extends through June 2026. Hull Data Ob... +2

This is a much stronger GeoSMILE application than classifying the crime type of an already-recorded event: it allows us to explain **why the model predicts different future crime levels between Hull neighbourhoods and how much of that prediction is attributable specifically to location**.

since now xall Feature-A branch: feature attribution-location excluded . Feature-B branch:feature attribution-location excluded

Correct. The key distinction is **not whether location is attributed**, because location is excluded from feature attribution in both branches. The distinction is whether location is available to the predictive model when estimating the effects of the non-spatial features.

Correct GeoSMILE structure

Branch	Model input	Variables receiving feature attribution	Purpose
Geo branch	Spatial coordinates/location	Location only	Measures geographical sensitivity and spatial contribution
Feature-A	Non-spatial features only	Non-spatial features	Explains features without spatial context
Feature-B	Non-spatial features + location	**Non-spatial features only; location excluded from attribution**	Explains how feature effects change when the model has spatial context

So for the Hull crime dataset:

```
Feature-A model inputs:
crime lags
demographics
deprivation indicators
built-environment variables
seasonality
```

```
Feature-B model inputs:
all Feature-A variables
+
LSOA centroid easting
LSOA centroid northing
```

But the Feature-B explanation vector remains:

```
Feature-B attributed variables:
crime lags
demographics
deprivation indicators
built-environment variables
seasonality
```

The coordinates are used by the Feature-B predictive model, but they are **not treated as ordinary features in its attribution output**. Their explanatory role is isolated in the Geo branch.

Therefore, the conceptual comparison is:

$$\text{Feature-A attribution} = \text{feature effects without location awareness}$$

$$\text{Feature-B attribution} = \text{feature effects after conditioning on location}$$

$$\text{Geo attribution} = \text{the separate effect or sensitivity associated with location}$$

For example, unemployment may receive a strong Feature-A attribution because it acts partly as a proxy for neighbourhood location. Its Feature-B attribution may become weaker once coordinates are available to the predictive model. The difference between the two branches indicates how much the apparent importance of unemployment depends on spatial context—not a direct attribution to the coordinates.

A precise label for the dashboard would be:

- **Feature-A: Feature attribution without spatial context**
- **Feature-B: Feature attribution conditioned on spatial context**
- **Geo branch: Location-specific sensitivity**

This is the architecture we should use when preparing the Hull dataset and GeoSMILE experiment.

ChatGPT can make mistakes. Check important info.